



Study of Leaching



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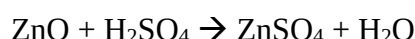
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Introduction:

Leaching is a simple process in extractive metallurgy through which a solid material—such as an ore or metal oxide—is attacked with a liquid solvent to selectively leach out specific constituents. During the process, the mineral dissolves in aqueous conditions, resulting either in total dissolution or release of some constituents along with the production of a solid residue known as gangue. The leaching is decided by the reagents used, common examples including acid leaching (utilizing acids like sulfuric acid), alkali digestion (utilizing strong bases like sodium hydroxide), and oxidation or reduction reaction leaching.

For instance, in acid leaching, zinc oxide reacts with sulfuric acid to form zinc sulfate and water, as shown by the reaction:



The speed and effectiveness of the leaching reaction are influenced by several parameters that are temperature, agitation rate, and concentration of the leaching reagent. Higher temperatures will accelerate the reaction, while agitation helps by reducing the boundary layer around particulate solids, thereby allowing faster diffusion of products and reactants. The kinetics of leaching are governed by processes such as diffusion of reagents to the mineral surface, surface chemical reaction, and diffusion of dissolved species from the surface.

The kinetic steps that determine the rate of leaching could be one or a combination of the following:

- 1) The diffusion of the reagent R from the bulk of the solution to the solid surface II.
- 2) The diffusion of the reagent R through the porous reaction product layer.
- 3) The reaction of the reagent with the mineral surface I to form a soluble metal species.
- 4) The diffusion of the product metal species M or other reaction products P through the porous reaction product layer in the outward direction.
- 5) The diffusion of the product metal species M (or other products P) away from the surface

Understanding these factors is essential for optimizing leaching processes in industrial applications, ensuring high recovery rates and efficient use of resources.

Apparatus:

- ZnO
- H_2SO_4
- Beakers
- Thermometer
- Hot plate, magnetic stirrer
- Electronic balance

Procedure:

- 1) Mix 10g ZnO with 200cc H_2SO_4 (10mL/L)
- 2) Then we placed the mixture in hot plate and maintained 50°C with no stirring
- 3) Then we cooled it for 5 minutes
- 4) We took 20 cc from the solution.
- 5) Then we added alkali, methyl orange to make the solution basic
- 6) Then, HCl was added
- 7) Then we added diammonium hydrogen phosphate which formed zinc pyrophosphate.
- 8) Then we washed and filtered the precipitate to extract Zn ppt.
- 9) Hot water is added continuously to maximize the extraction of Zn ppt.
- 10) Pulp is burned along with pulp ash, and $\text{Zn}_2\text{P}_2\text{O}_7$ remains in the crucible.

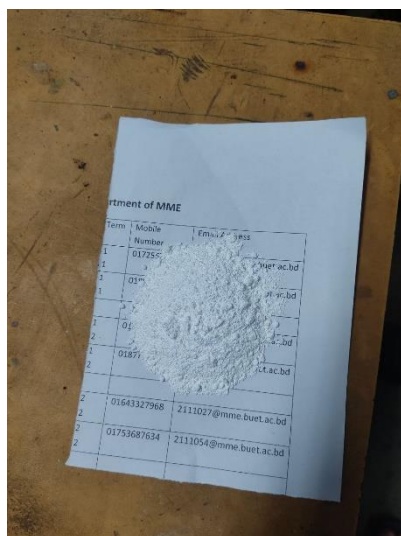


Figure 1: 10g ZnO



Figure 2: Maintaining temperature at 50°C



Figure 3: Filtering through pulp

Calculation and Result:

Weight of empty crucible = 30.61 gm

Weight of crucible with ppt. = 30.94 gm

Weight of $\text{Zn}_2\text{P}_2\text{O}_7$ = (30.94-30.61-0.0014) gm = 0.3286 gm

[$\text{Zn}_2\text{P}_2\text{O}_7$ has 42.9%Zn]

Zn content in 20 cc solution = $0.3286 \times 0.429 = 0.141$

Zn content in 200 cc solution = $0.141 \times 10 = 1.41$ gm

$\% \text{Zn} = \frac{1.41}{10} \times 100 = 14.1\%$

Discussion:

The leaching of zinc oxide with sulfuric acid was demonstrated successfully in the experiment, with zinc recovery being 14.1%. Gravimetric analysis for zinc pyrophosphate, a satisfactory technique for zinc determination in solution, was employed to ascertain this value. The recovery obtained is likely lower than what it could have been, owing to under dissolution of zinc oxide or loss during filtration and washing.

Kinetics of leaching has 5 steps and which are :

1. The diffusion of the reagent R from the bulk of the solution to the solid surface II.
2. The diffusion of the reagent R through the porous reaction product layer.
3. The reaction of the reagent with the mineral surface I to form a soluble metal species.
4. The diffusion of the product metal species M or other reaction products P through the porous reaction product layer in the outward direction.
5. The diffusion of the product metal species M (or other products P) away from the surface.

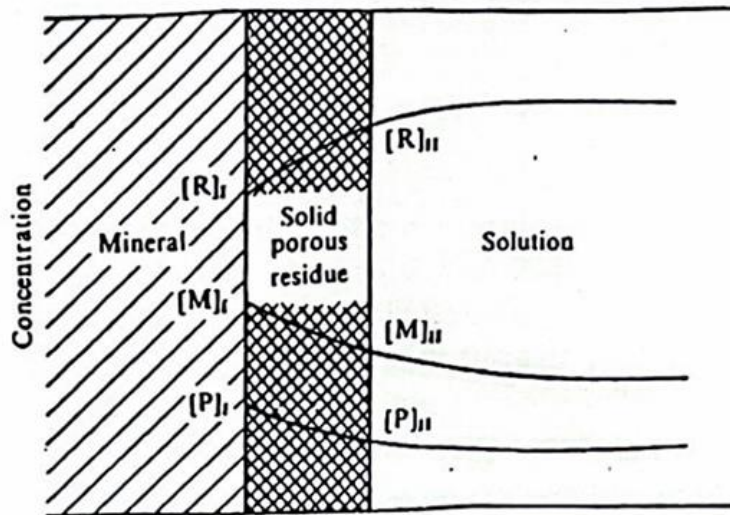


Figure 4: Representation of Mineral Surface Decomposing & Dissolving

But in our lab experiment, we took pure ZnO sample (lab graded); not any ore or mineral. So there less likely to arrive gangue/ any other porous residue or the amount is so small that it could be neglected.

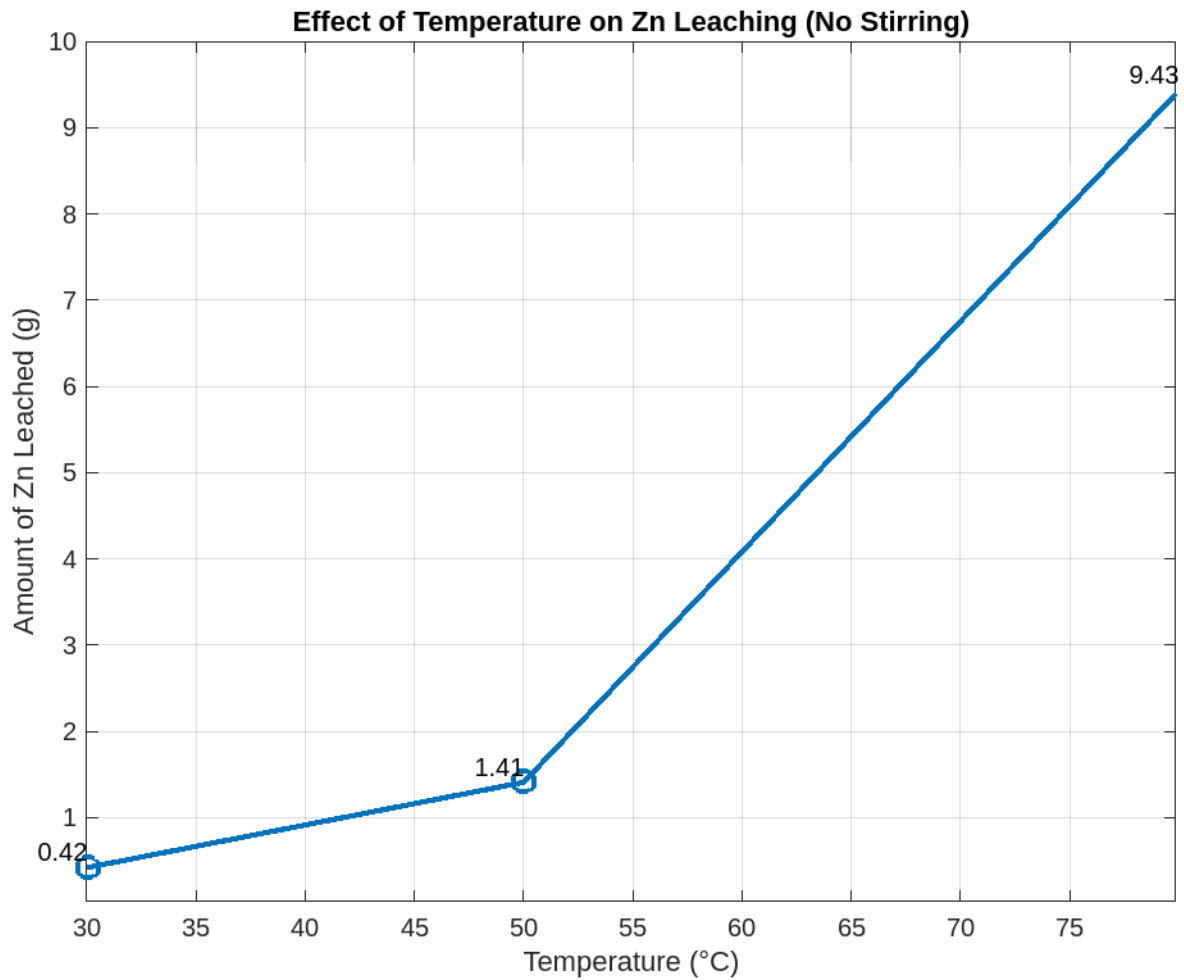
As a result step 2 and 4 in kinetics of leaching in our experiment can be neglected .

In addition if step 1 and 5 are rate controlling, agitation can enhance the process and if step 3 is rate controlling , temperature can enhance the process.

Amount of Zn in leaching solⁿ of all group is given below:

No Stirring		
Group	Parameters (Temperature)	Amount of Zn (200cc or 10g)
1	80°C	9.43
2	Room temperature (30°C)	0.42
3	50°C	1.41
Room Temperature		
Group	Parameters (stirring)	Amount of Zn (200cc or 10g)
4	mild	1.37
5	No	0.89
6	High	1.37

Generated graph (using matlab):

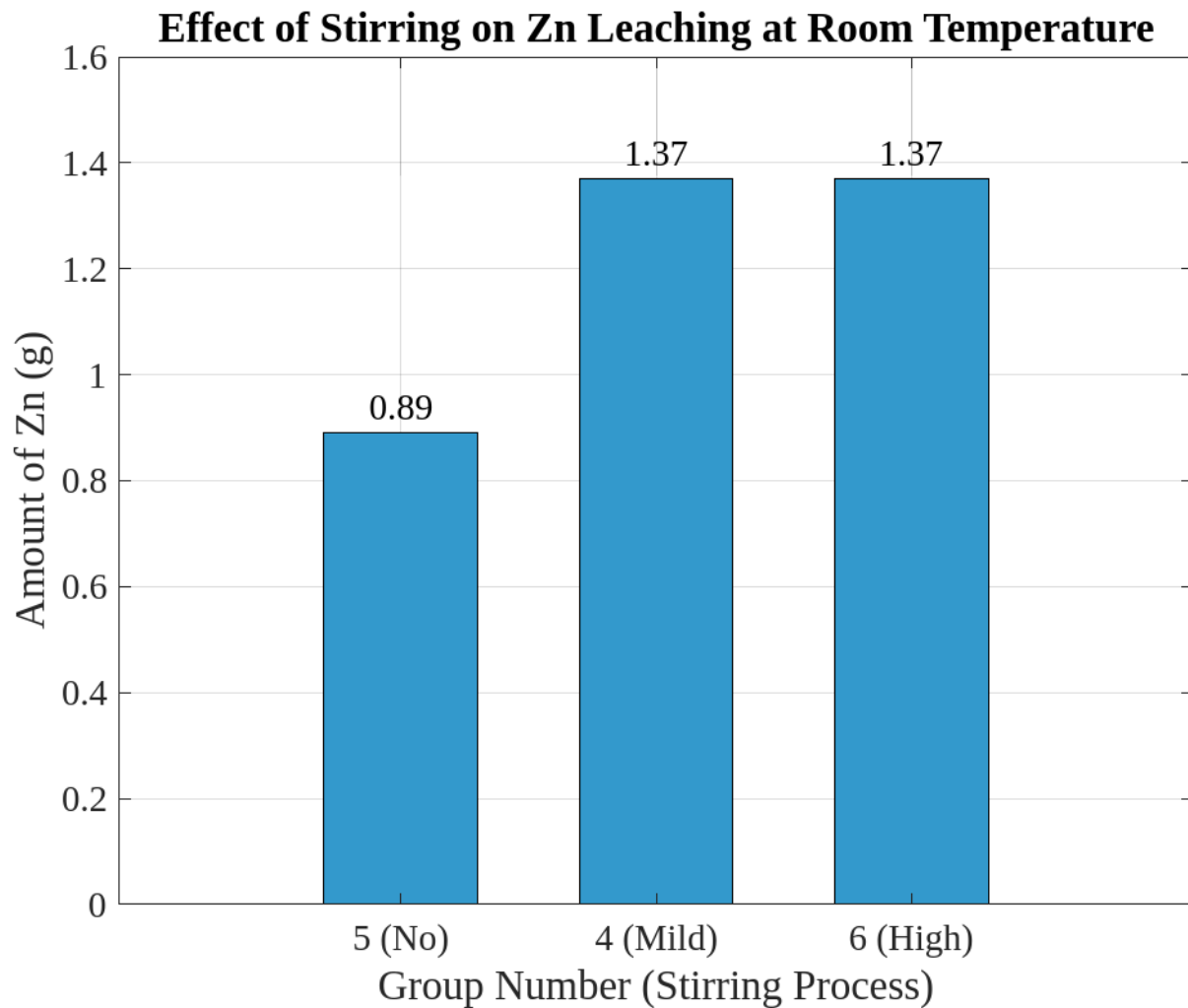


It is apparent from the graph and data that **zinc recovery is significantly increased with temperature without stirring**. Zinc recovery reached a maximum of 9.43g at 80°C. while decreasing sharply to as little as 0.42g at room temperature.

Explanation

Higher temperatures enhance the leaching kinetics by increasing the rate of the ZnO–acid interface reaction. This aligns with step 3 of the leaching kinetics (the surface chemical reaction), which is also found to be temperature-dependent. The Arrhenius equation supports this in that higher thermal energy overcomes activation barriers for a more rapid dissolution of ZnO into Zn^{2+} ions.

$$\ln K = \ln A - \Delta E / (RT),$$



Under room temperature, stirring exhibited moderate improvement of zinc recovery. Slight and vigorous agitation yielded the same quantity of zinc (1.37g) which was considerably higher than that recovered in the absence of stirring (0.89g).

Explanation:

Stirring reduces the thickness of the boundary layer surrounding the ZnO particles and hence enhances mass transfer. This is related to step 1 and step 5 in leaching kinetics:

Step 1: Diffusion of acid (H_2SO_4) to the surface of ZnO.

Step 5: Diffusion of dissolved Zn^{2+} ions from the reaction site.

Both these diffusion steps are facilitated under stirring, thereby allowing more zinc to enter the solution phase.

Interestingly, **no further augmentation with increasing stirring intensity over a gentle condition was produced, presumably due to diffusion control saturation or room temperature equilibrium constraints.**

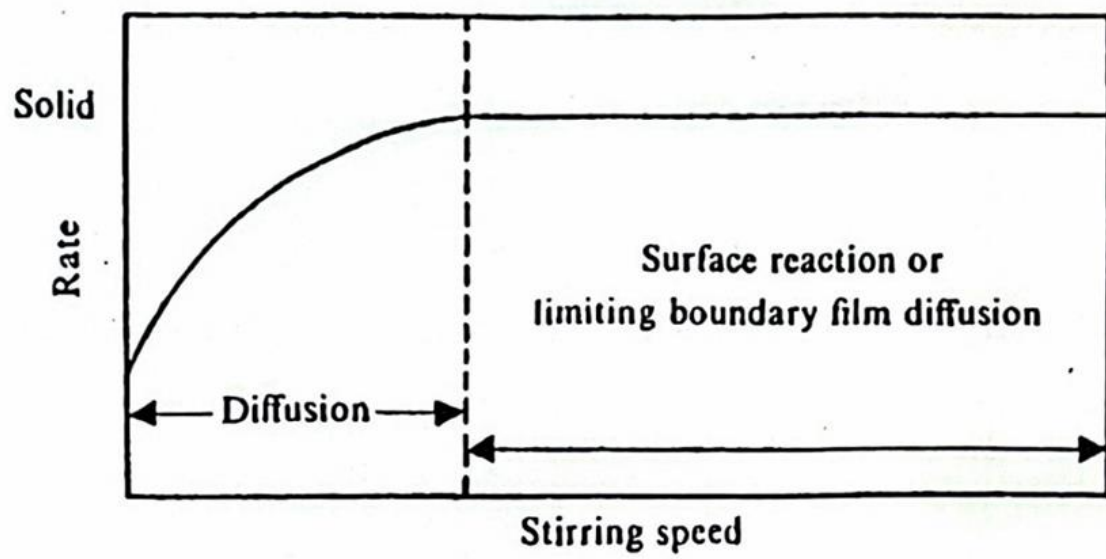


Figure 5: Reaction rate versus stirring speed